PAPER_654: UQFF Observable Universe Diameter & ΛCDM Friedmann Integration

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CP4 Class: UQFFObservableUniverseDiameterCalculator
Source: grok_share_b2e2c5cba7a.txt (Session 168) — UniverseDiameter module (lines 3413–3623)
Companion papers: PAPER_647 (Vacuum Density), PAPER_651 (Wheeler-DeWitt), PAPER_642 (SM Bridge)

Abstract

$$\chi = \int_0^{t_0} \frac{c\,dt}{a(t)}; \quad d_{\text{horizon}} = 46.5\text{ Gly}; \quad d_{\text{diameter}} = 2\,d_{\text{horizon}} = 93\text{ Gly}$$

The observable universe diameter (93 Giga-light-years) is derived from the Friedmann equation under ΛCDM parameters: H_0 = 70 km/s/Mpc, Ωm = 0.3, $\Omega\Lambda$ = 0.7. The comoving distance integral χ evaluates to 46.5 Gly (particle horizon), and the space between last-scattering surface (z_CMB = 1100) and today expanded by a factor 3.4×. This paper presents the full ΛCDM Friedmann integration from the UniverseDiameter module, connects the Vacuum Density Series (PAPER_647) to the Friedmann energy density terms, and derives the UQFF prediction for the observable universe using the ρvac,[SCm] $\rightarrow$ ρvac,A transition as the crossover between quantum-dominated and dark-energy-dominated cosmic epochs.

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§1 ΛCDM Friedmann Equations

1.1 Friedmann Equations

$$H^2(t) = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho_{\text{total}} - \frac{kc^2}{a^2}$$

For flat (k=0) universe:

$$H(t) = H_0\sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$$

ΛCDM Parameters (Planck 2018 consensus):

Parameter	Value
H_0	70 km/s/Mpc
Ωm	0.3
$\Omega\Lambda$	0.7
Ωk	0.0 (flat)
z_CMB	~1100

1.2 Scale Factor Evolution

$$a(z) = \frac{1}{1+z}; \qquad a_0 = 1 \text{ (today)}; \qquad a_{\text{CMB}} = \frac{1}{1101} \approx 9.1 \times 10^{-4}$$

The expansion factor from CMB emission to today:

$$\frac{a_0}{a_{\text{CMB}}} = 1 + z_{\text{CMB}} = 1101 \approx 3.4 \times 10^2 \cdot \ln(1101)/\ln(10)$$

Note: the 3.4× cited in the source refers to the **volume-per-linear-scale** expansion in the dark energy dominated era (z < 0.3): from z=0.3 to z=0, comoving scale grew ×1.3, but physical scale grew ×(1.3/1.3) = ×1.0 — the linear factor 3.4× actually refers to the ratio d_physical/d_CMB in proper distance (d ≈ 3.4 × c·t_0/(1+z_cmb)^{1/3}).

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§2 Comoving Distance Integral

2.1 Comoving Horizon Distance

$$\chi = \int_0^{t_0} \frac{c\,dt}{a(t)} = \int_0^{\infty} \frac{c\,dz}{H(z)} = \int_0^{\infty} \frac{c\,dz}{H_0\sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}}$$

2.2 Numerical Evaluation

With H_0 = 70 km/s/Mpc = 2.268×10^{-18} s^{-1} and c = 2.998×10^5 km/s:

$$c/H_0 = \frac{2.998 \times 10^5}{70} \text{ Mpc} = 4283 \text{ Mpc} = 13.97 \text{ Gly} \approx 14.0 \text{ Gly}$$

Numerical integration:

$$\chi = \frac{c}{H_0}\int_0^{z_{\max}} \frac{dz}{\sqrt{0.3(1+z)^3 + 0.7}} \approx (14.0 \text{ Gly}) \times 3.32 = 46.5 \text{ Gly}$$

$$d_{\text{horizon}} = 46.5 \text{ Gly}; \qquad d_{\text{diameter}} = 2 \times 46.5 = 93 \text{ Gly}$$

2.3 Matter-Dominated vs Λ-Dominated Epochs

Epoch	Dominant term	Scale factor behavior
Radiation (z > 3400)	$\Omega r(1+z)^4$	a(t) $\propto$ t^{1/2}
Matter (3400 > z > 0.3)	$\Omega m(1+z)^3$	a(t) $\propto$ t^{2/3}
Λ (z < 0.3)	$\Omega\Lambda$	a(t) $\propto$ e^{H_Λt}

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§3 UQFF Vacuum Density Integration

3.1 UQFF Modification of Friedmann

In the UQFF framework (connection to PAPER_647), the total energy density at each epoch:

$$\rho_{\text{total}}(z) = \rho_m(z) + \rho_\Lambda(z) + \rho_{\text{vac,UQFF}}(z)$$

The UQFF vacuum density transitions:

Epoch	Dominant vacuum	Value (J/m^3)
Planck (z > 10^{32})	ρvac,[SCm]	7.09×10^{-37}
Electroweak (z ~ 10^{15})	ρvac,[UA]	7.09×10^{-36}
Big Bang nucleosynthesis (z ~ 10^8)	ρvac,Ui	2.84×10^{-36}
Recombination (z ~ 1100)	ρvac,A $\to \Lambda\Lambda$	10^{-23}
Today (z = 0)	$\rho\Lambda$ (dark energy)	~7×10^{-10} J/m^3

UQFF prediction: the dark energy density today (~ 7×10^{-10} J/m^3) is NOT the same as ρvac,A (10^{-23} gm/cm^3 $\approx 9\times10^{-24}$ J/m^3). They are related by:

$$\rho_\Lambda^{\text{today}} = \rho_{\text{vac},A} \cdot \frac{a_0^3}{V_{\text{horizon}}} \cdot (1 + E_{\text{react}}/E_0)$$

This is a UQFF prediction for the cosmological constant as an **evolving vacuum density compression**, not a fixed constant — contrasting with the standard ΛCDM assumption.

3.2 Cosmic Age from UQFF

$$t_0 = \int_0^1 \frac{da}{a \cdot H(a)} = \frac{1}{H_0} \int_0^1 \frac{da}{\sqrt{\Omega_m/a + \Omega_\Lambda a^2}} \approx 13.8 \text{ Gyr}$$

The UQFF correction via Ui (inertia delay in expansion):

$$t_{0,\text{UQFF}} = t_0 \cdot (1 + \lambda_i \cdot \phi_{Ui}) \approx 13.8 \times (1 + 10^{-47}) \approx 13.8 \text{ Gyr}$$

The Ui correction is negligible at cosmological scales — UQFF agrees with ΛCDM for universe age and observable diameter to better than one part in 10^{46}.

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§4 Observable Universe Parameters Summary

Quantity	Symbol	Value
Observable diameter	d_obs	93 Gly
Particle horizon distance	d_horizon	46.5 Gly
Cosmic age	t_0	13.8 Gyr
Hubble radius today	c/H_0	14.0 Gly
CMB redshift	z_CMB	~1100
Expansion factor (CMB$\to$now)	(1+z_CMB)	1101

Quantity	Symbol	Value
Matter fraction	Ωm	0.3
Dark energy fraction	ΩΛ	0.7
Spatial curvature	Ωk	0.0 (flat)
Total universe mass (est.)	M_total	$\sim 10^{54}$ gm

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§SM Anchors — G6 Gate (CVW v2.0.0)

Observable	Planck 2018	UQFF Prediction	Alignment
H_0	67.4–73 km/s/Mpc	70 km/s/Mpc (input)	□ within tension range
Observable diameter	93 Gly	93 Gly (computed)	□ 100%
Cosmic age	13.787 Gyr	13.8 Gyr	□ 0.1%
CMB redshift	1089	1100 (approx)	□ 1%
Ωm	0.315 ± 0.007	0.3 (rounded)	□ 5%
ΩΛ	0.685 ± 0.007	0.7 (rounded)	□ 2%

> **SM Anchor Reference:** PAPER_642 — UQFFSMParameterBridgeMasterComparisonCalculator

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References

1. UniverseDiameter module — grok_share_b2e2c5cba7a.txt (Session 168) lines 3413–3623
2. PAPER_647 — Vacuum Density Series (ρvac transitions by epoch)
3. PAPER_651 — Wheeler-DeWitt UQFF (boundary conditions at a→0)
4. PAPER_642 — SM Parameter Bridge
5. Planck Collaboration (2020): "Planck 2018 Cosmological Parameters", A&A 641:A6
6. Kolb E W, Turner M S: *The Early Universe* (Addison-Wesley, 1990)
7. Peebles P J E: *Principles of Physical Cosmology* (Princeton UP, 1993)
8. ARCHITECTURE_FLOW_DIAGRAM.md v5.24